might guess, there is nothing magic about the number two. The ideas presented here generalize to larger systems that require many operators to characterize them.

To work with two different compatible operators, we'll need two sets of labels for their basis vectors. We'll use the labels λ_i and μ_a. The symbols λ_i and μ_a are the eigenvalues of **L** and **M**. The subscripts *i* and *a* run over all the possible outcomes of measurements of **L** and **M**. We assume that there is a basis of state-vectors $|\lambda_i, \mu_a\rangle$ that are simultaneous eigenvectors of both observables. In other words,

$$\mathbf{L}|\lambda_i, \mu_a\rangle = \lambda_i|\lambda_i, \mu_a\rangle$$

$$\mathbf{M}|\lambda_i, \mu_a\rangle = \mu_a|\lambda_i, \mu_a\rangle.$$

To make these equations a little less precise but a little easier to read, I will sometimes leave out the subscripts:

$$\mathbf{L}|\lambda, \mu\rangle = \lambda|\lambda, \mu\rangle$$

$$\mathbf{M}|\lambda, \mu\rangle = \mu|\lambda, \mu\rangle.$$

In order to have a basis of simultaneous eigenvectors, the operators **L** and **M** must commute. This is easy to see. We begin by acting on any of the basis vectors with the product **LM**, and then use the fact that the basis vector is an eigenvector of both:

$$\mathbf{LM}|\lambda, \mu\rangle = \mathbf{L}\mu|\lambda, \mu\rangle,$$

or